Curve Dashboard Build Plan

What we will build for the crude-quant-dashboard, in build order, and how each piece helps find mispricing in WTI, Brent and WTI-Brent flies and double-flies

crude-quant-dashboard · October 2026 · Planning document, nothing here is built yet

1. The approach in one page

A **mispricing** is a structure (fly, double-fly, intermarket fly or a custom combination) that sits away from fair value with no reason to be there. No single fair value is reliable, so the dashboard compares every structure against **five independent references**, then passes it through **four filters**. A structure is flagged only when several references agree and the filters pass.

Reference	Question it answers	Built by component
Neighbours	Is this point out of line with the structures next to it on the curve?	4 Structure calculator
Own history, age-matched	Is it extreme for this structure at this distance from expiry?	9 Age-matched history
Model	Is it off the fitted fair curve?	5 PCA, 6 Nelson-Siegel
Cross-market	Did Brent move here but WTI didn't, or the reverse?	4, 11 Fly curve overlay
Beta-adjusted move	Did it move more or less than the liquid curve says it should?	7 Beta engine
Seasonal (extra)	Is it away from where it usually is on this date?	10 Seasonality tool

Filter	What it rules out	Built by component
Reason check	Kinks with a known cause: expiry, index roll, December contracts, events, known hedges	13 Event calendar
Tradability	Mispricings too small to pay for crossing four bid-ask spreads	14 Cost filter
Regime	Structures that are still trending away	8 Mean-reversion stats
Persistence	Kinks that have sat unchanged for weeks (likely structural)	12 Rich/cheap board

> **DIVISION OF LABOUR**
> The dashboard finds, ranks and explains opportunities. Execution stays in TT. The dashboard does not route orders.

2. What we have to build with

Source	What it gives
Reuters hourly CSVs (5 years)	Per named contract: **Close Bid, Close Ask, Last, Close Bid/Ask Size, Volume**. Open/High/Low are kept in the raw files but not loaded
Live feed (Lightstreamer) + structure engine	Real-time leg prices and live pricing of flies, double-flies and custom structures. Already running
QH Data API	Curves endpoint (daily curve data). OHLC endpoints are not entitled

Postgres (existing)	Users, teams, books, positions, trade journal. New market-data tables go here too
Frontend (existing)	Curves tab with curve tiles and a strategy builder, positions tab, journal tab. Built on tradecn/shadcn

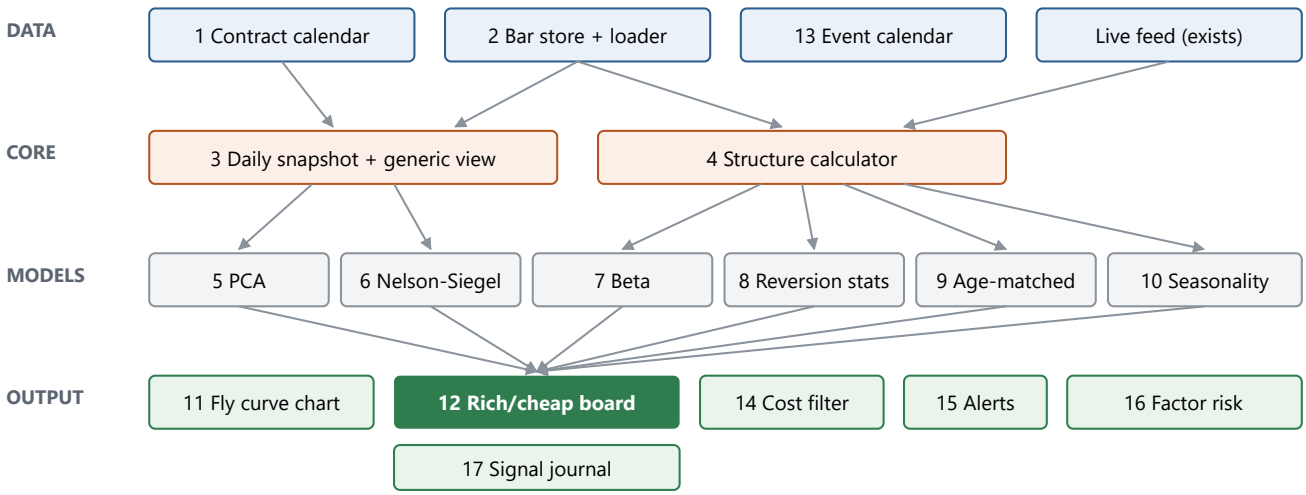


Figure 1. Build layers. Each layer depends on the one above it. All models feed the rich/cheap board (12), along with the event calendar (13) and cost filter (14). The board is where every reference and filter comes together.

3. Phase A: data foundations

Nothing in this phase finds a mispricing by itself, but every later reference is only as good as these pieces. Most false signals in curve trading come from bad data, not bad models.

1 Contract calendar

POSTGRES TABLE

WHAT	One row per named contract (CLZ26, COF27…) with product, delivery month, last trading day and, for WTI, first notice day. Includes ICE and CME holiday calendars.
BUILT FROM	Contract codes in the Reuters CSVs, plus exchange expiry rules: Brent expires on the last business day of the second month before delivery; WTI three business days before the 25th of the month before delivery.
HOW IT WORKS	The loader creates a row the first time it sees a contract. Expiry dates are computed from the rules and holidays, then checked against the exchange calendar.
MISPRICING ROLE	Gives every structure its **time to expiry**, which age-matched history, Nelson-Siegel, the seasonality tool and the expiry reason flags all need. It also lets the system tell an expiry distortion from a real mispricing.

2 Hourly bar store and CSV loader

POSTGRES TABLE + PYTHON SCRIPT

WHAT

Table `bars_1h`: contract, bar-end time in UTC, Last, Close Bid, Close Ask, Close Bid/Ask Size, Volume. Raw CSVs archived compressed outside OneDrive.

HOW IT WORKS

- Normalises timestamps to UTC with a consistent bar-end convention, handling daylight saving.
- Bulk-loads into a staging table, then inserts with "skip duplicates", so re-loading a file is harmless.
- Stores a missing or one-sided quote as empty (NULL), never as zero.
- Prints a load report: rows loaded, duplicates, one-sided quotes and gaps per contract.

MISPRICING ROLE

Close Bid and Close Ask give a **synchronised mid** for every leg at every hour, whether or not it traded. This removes the main source of false double-fly signals in back months: one leg showing an old traded price.

3 Daily snapshot and generic-position view

POSTGRES VIEWS

WHAT

- **Daily snapshot:** each contract's mid at one fixed New York time (e.g. the bar ending 15:00), the same moment for WTI and Brent.
- **Generic view:** maps each named contract to c1, c2, c3… on each date using a roll rule we can change without reloading data.

HOW IT WORKS

Both are views (or small tables refreshed after each load) built from tables 1 and 2. Daily changes are always computed **within the same named contract**, then mapped to its generic position, so rolls never create fake jumps.

MISPRICING ROLE

The clean daily curve that PCA, Nelson-Siegel, betas and history all run on. Synchronised WTI and Brent snapshots make intermarket comparisons valid.

4 Structure calculator

BACKEND SERVICE + API

WHAT	Computes any structure's history from leg prices: spreads, flies, double-flies, dfly-vs-dfly, intermarket versions and custom weights from the strategy builder.
HOW IT WORKS	• **Mid:** Σ weight × leg mid. • **Bid/ask:** to buy, pay the ask on positive-weight legs and receive the bid on negative-weight legs; selling is the reverse. This is an upper bound on cost, since listed spread markets are usually tighter. • **Validity flag:** a value counts only when every leg has a two-sided quote within a maximum bid-ask width. • **Factor readout** of the weights: level, slope and curvature exposure (e.g. Fly(G) − 2·Fly(H) is level- and slope-neutral but keeps a short fly). • Uses the same weights logic as the existing live structure engine, so historical and live values match.
MISPRICING ROLE	**The neighbours reference.** A double-fly is literally the size of the kink between two adjacent flies, and dfly-vs-dfly pinpoints exactly where on the curve the kink is. It also provides the cross-market series (WTI, Brent, arb) and the bid/ask needed by the cost filter.

4. Phase B: fair-value models and statistics

Each component here produces one fair-value reference: a number saying how far a structure is from where it "should" be, in standard deviations.

5 PCA curve model

BACKEND JOB, DAILY

WHAT	Principal component analysis of daily curve changes, separately for WTI, Brent and the arb, over a rolling 1–3 year window.
HOW IT WORKS	• Input: daily same-contract changes for positions c1…cN (component 3), with the front month excluded or flagged in its last days before expiry. • The first three components are level, slope and curvature. Together they usually explain over 95% of daily curve moves. • Rebuild each structure from those three components. The difference between actual and rebuilt is the **residual**, and its z-score is the signal. • Also gives the factor loadings used to hedge and to measure position risk (component 16).
MISPRICING ROLE	**Model reference.** A fly is mostly curvature plus noise; a double-fly is almost entirely residual. A large residual z-score means the structure has moved in a way the curve's normal behaviour does not explain, which is exactly the kind of move that tends to revert.

6 Nelson-Siegel / Svensson curve fit

BACKEND JOB, DAILY

WHAT	A smooth parametric curve fitted to each day's prices against time to expiry, one fit per market.
HOW IT WORKS	• Fixed decay parameter, so the level, slope and curvature coefficients form stable time series. • Svensson's extra term handles crude curves that bend near the front (backwardated front, contango back). • The fitted curve gives a fair price for every contract; fair flies follow directly. • **Roll-down:** the fitted curve also says how a structure's value changes as it ages, which gives its carry.
MISPRICING ROLE	**Second model reference**, built differently from PCA: PCA asks "is this move normal?", Nelson-Siegel asks "is this price on a smooth curve?". When both flag the same structure, the signal is much stronger. It also draws the **fair band** on the fly curve chart.

7 Beta engine

BACKEND JOB, DAILY + LIVE

WHAT	For each structure, its beta to flat price (front-month mid) and to the front spread, estimated on hourly mid changes over a rolling window with a robust regression.
HOW IT WORKS	Live: expected move = β × today's change in flat price or front spread. Compare with the actual move. The gap is the **beta-adjusted residual**.
MISPRICING ROLE	**Beta-adjusted reference.** Catches structures that lagged (or overshot) what the liquid part of the curve just did. It also cleans the other references: a fly that looks cheap only because price fell is not a mispricing.

8 Mean-reversion statistics

BACKEND, PER STRUCTURE

WHAT	Rolling z-score, Bollinger position, RSI, Ornstein-Uhlenbeck **half-life** (how fast it reverts), Hurst exponent, and a close-only trend measure (ADX-style).
HOW IT WORKS	Computed on the structure's mid series from component 4, using only valid bars.
MISPRICING ROLE	**Regime filter and timing.** Hurst below 0.5 and a short half-life mean the structure has historically reverted; a strong trend reading says "don't fade this yet". Half-life also sets how long to hold and when a trade has failed.

9 Age-matched history

BACKEND, PER STRUCTURE

WHAT	For a structure today, its percentile against the **same structure at the same number of days to expiry** in past years (e.g. this year's G-H-J fly at 120 days vs last year's at 120 days).
HOW IT WORKS	Five years gives about five past observations per point, which is thin. Optionally pool neighbouring structures (one month earlier or later) to enlarge the sample, clearly labelled as pooled.
MISPRICING ROLE	**History reference.** Comparing against the same age avoids a common trap: structures behave differently as expiry approaches, so a plain calendar-window z-score would flag normal ageing as a mispricing.

10 Seasonality tool

NEW DASHBOARD TAB

WHAT	An improved version of the existing seasonality display: any structure (chain or custom weights) plotted for each past year alongside the current one.
IMPROVEMENTS	• Median path with 25–75% and 10–90% bands instead of loose lines, plus today's percentile for this date. • Zero-based paths by default, with a chosen anchor date. • Alignment by days to expiry, with each leg's expiry marked and the index-roll window shaded; option to cut series N days before the first expiry. • **Entry/exit window finder:** per-year P&L, hit rate and worst drawdown for a chosen window, plus a heatmap over all entry and exit dates. • Years tagged by regime (backwardation/contango, front-spread level) with a "years like this one" filter. • Strip view: the same structure for every start month as small multiples. • Outlier days marked with a reason instead of silently removed.
MISPRICING ROLE	**Seasonal reference.** Today's distance from the seasonal median for this date. It also works as a reason check in reverse: a kink that appears every year at this point is probably seasonal, not mispriced.

5. Phase C: the mispricing indicator

This phase turns the references into one ranked, explained view the trader can act on.

11 Fly curve chart with fair band and overlays

EXTENDS THE CURVES TAB

WHAT	The existing fly-curve tile, extended with: the Nelson-Siegel fair curve and a ±1σ band; today vs previous settle vs one week ago; and a WTI / Brent / arb overlay on one chart. Same view for double-flies.
MISPRICING ROLE	Makes kinks **visible**: a point outside the band is off the model; a point whose WTI and Brent versions diverge is a **cross-market** signal. This is the trader's sanity check on what the board ranks.

12 Rich/cheap board

NEW DASHBOARD TAB, THE CORE

WHAT	A heatmap: rows are structure types (fly, dfly, intermarket fly, intermarket dfly) for WTI, Brent and the arb; columns are contract months; colour is the composite score. Clicking a cell opens a detail row and the structure's charts.
DETAIL COLUMNS	Z-score from each reference and how many agree; expected reversion in ticks; current bid-ask width and reversion-to-cost ratio; half-life; dislocation age (days since it opened); reason flags.
HOW THE SCORE WORKS	• Each reference gives a signed z-score (positive = rich, negative = cheap). • **Composite** = weighted average of the z-scores, shown together with the **agreement count** (how many references point the same way). • Starting weights are equal. The signal journal (17) later shows which references actually predicted reversion, and the weights are adjusted from that evidence. • Filters dim or hide a cell rather than changing its score, so the trader can always see why something was excluded.
MISPRICING ROLE	**This is the mispricing indicator.** One reference alone fires on structural kinks; three or more agreeing, with no reason flag, enough edge over cost and no trend, is a candidate trade. Dislocation age catches persistent kinks that are probably structural.

13 Event and reason-flag calendar

POSTGRES TABLE + FLAGS

WHAT	Dated events: expiries and options expiries for both markets (from component 1), the index-roll window (business days 5–9), EIA and API reports, OPEC+ meetings, month-end. Plus static flags: December contracts, front months near expiry.
MISPRICING ROLE	**Reason check.** Attaches the likely explanation to a kink before it is traded. A kink with a known cause can persist or grow, so flagged structures are dimmed on the board, not removed.

14 Cost and tradability filter

BACKEND + BOARD COLUMN

WHAT	For each structure: round-trip cost from the leg bid/ask (component 4), live from the feed and historically from the Reuters close bid/ask; available size from the bid/ask sizes.
HOW IT WORKS	Expected reversion (distance to fair value) divided by round-trip cost. Require at least 2–3×. Because leg-built cost is an upper bound, structures that just miss can be checked against the listed spread market in TT.
MISPRICING ROLE	**Tradability filter.** Separates real opportunities from mispricings that exist only on paper, which is common in back-month double-flies with four wide legs.

15 Alerts

DASHBOARD NOTIFICATIONS

WHAT	A notification when a structure crosses the composite threshold **and** passes tradability **and** has no reason flag **and** isn't trending.
MISPRICING ROLE	Lets one trader watch the whole curve across three markets. Thresholds are kept strict on purpose: too many alerts and they get ignored.

16 Factor-exposure risk panel

EXTENDS THE POSITIONS TAB

WHAT	Net level, slope and curvature exposure (PCA loadings from component 5) and net flat-price beta (component 7) across the open positions already stored in the positions table.
MISPRICING ROLE	Makes sure a book of mispricing trades stays a **curvature** bet. Without it, several flies that each look neutral can add up to an accidental slope or level position, and the P&L then depends on price direction rather than the mispricing closing.

17 Signal journal

EXTENDS THE JOURNAL TAB

WHAT	Every board flag is logged with a snapshot of its references, score and filters, whether it was traded, and what the structure did over the following days. Traded flags link to the existing trade journal entries.
HOW IT WORKS	Analytics on top: hit rate and average outcome by reference, by agreement count, by structure type, by market and by reason flag.
MISPRICING ROLE	**Calibrates the indicator.** After a few months it shows which references and filters actually predict reversion in these markets, and feeds the board's weights. This feedback loop is the main advantage of a home-built tool over a screen.

6. Phase D: later, once more data is available

Addition	Needs	What it adds
Live bid/ask recorder	Existing live feed; just store snapshots every minute or so	True intraday costs and minute-level history going forward
Daily settlement + open interest	Small daily file per contract	Exact settles for the models; open-interest changes reveal where hedgers are building positions (flow detection)
Listed spread instruments	Reuters data for exchange-listed calendar spreads	The true cost of the most-traded structures, tighter than legging
Intraday lead-lag	Minute data (from the recorder)	Session leadership between WTI and Brent; entry timing
Fundamentals overlay	EIA weekly data, North Sea loading programmes	Context for front-spread moves; better reason checks

7. Build order and what each step unlocks

Step	Components	Usable result after this step
A	1 Contract calendar, 2 Bar store + loader, 3 Snapshot + generic view, 4 Structure calculator	Five years of clean history for any structure, viewable on demand. Already useful for manual analysis
B1	8 Reversion stats, 9 Age-matched history, 10 Seasonality tool	First mispricing signals from history alone, plus the seasonality tool replacing the current one
B2	5 PCA, 6 Nelson-Siegel, 7 Beta	Model and beta references; the fair band on the fly curve
C1	11 Fly curve upgrade, 13 Event calendar, 14 Cost filter	Visual kink detection with reasons and costs attached
C2	12 Rich/cheap board, 15 Alerts	The full mispricing indicator
C3	16 Factor risk, 17 Signal journal	Risk control and the feedback loop that calibrates the board

OPEN DECISIONS

- Daily snapshot time (proposed: the bar ending 15:00 New York, just after both settlement windows).
- Roll rule for the generic positions (proposed: roll a fixed number of business days before expiry).
- Default alignment for the seasonality tool: calendar date or days to expiry (proposed: days to expiry).
- How far out the curve analysis goes (depends on how many contracts the Reuters files cover).

Contract rules (last trading days, settlement windows) to be confirmed against current ICE and CME rulebooks before they are coded. This document is a planning aid, not trading advice.